

GeoMatch: photo geolocation as visual retrieval

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1 Why retrieval

We predict by **retrieval**: embed query and bank, copy the nearest match’s coordinate. Retrieval is not continuous — its output is restricted to the 11,758 training coordinates — but that discretisation is cheap: the geographically nearest training photo is a median 3.29 km away and within 50 km for 99.8% of queries, far below the error we achieve, whereas a geo-cell classifier is bounded by its cell size. The gain is that the encoder never regresses a coordinate; it only learns which photos were taken near each other, which supervises densely from every pair in a batch.

We did build the geo-cell alternative first: the same trunk with six hierarchical classification heads, decoded by a combinatorial partition decoder. It reached 75.2 km and we dropped it for the retrieval line, which was *worse* at that point (89.2 km) but had the lower floor and, once self-supervised pretraining was added, overtook it decisively. Table 1 keeps the non-monotonic ordering rather than hiding it.

2 Model (Fig. 1)

One RegNetY-400MF trunk [1] (`weights=None`) is shared across three views: the full frame at 512^2 and aligned left/right crops at 256^2 , which keep detail a single down-scaled frame destroys.

Global branch. Multi-scale GeM pooling [2] over stages 2–4, fused across views, through an identity-initialised 384×384 layer, L2-normalised to a descriptor g . **Local branch.** Stage-4 features average-pooled to 4×4 per view (48 tokens), projected $440 \rightarrow 64$, L2-normalised to tokens T .

The shortlist is the cosine top-200 of $\langle g_q, g_b \rangle$, rescored by a symmetric MaxSim late interaction [3]:

$$s(q, b) = 0.2 \langle g_q, g_b \rangle + 0.8 \cdot \frac{1}{2} \left(\overline{\max}_j T_q T_b^\top + \overline{\max}_i T_q T_b^\top \right)$$

The top-1 candidate’s coordinate is the prediction.

3 Training

Evaluated on the official 5-fold split: train on four folds, predict the fifth, fixed epoch count, no checkpoint selection.

(1) **BYOL pretraining** [4] of the trunk, labels unused, 160 epochs. (2) **Country-aware retrieval fine-tuning**, 40 epochs, AdamW (8×10^{-5} trunk, 6×10^{-4} heads), physical batch 32 with 2 gradient-accumulation steps (effective 64), EMA 0.999. Each group is an anchor, two geographic positives and one *same-country* hard negative 60–700 km away, re-mined from the current EMA encoder every epoch. Positives are the 8 most descriptor-similar of the 64 geographically nearest that lie within

50 km; for the 10.7% of anchors with fewer than 8 such neighbours the sampler falls back to the 8 nearest *regardless of distance*, so “ ≤ 50 km” holds for $\approx 89\%$ of anchors, not all. Anchors from the weakest countries are oversampled (DE 3.0 $\times$, FR 2.8, PL 2.0, IT/ES 1.7, SE 1.4, GB 1.3). Objective: 0.65 retrieval + 0.2 classification regulariser.

4 Results

pooled 5-fold OOF	runs	med.	mean	<200	<750
Geo-cell classification (dropped)	1	75.2	—	—	—
From-scratch retrieval baseline	1	89.2	618	55.9%	69.4%
+ BYOL + country-aware, 22 ep	1	58.0	490	60.7%	76.2%
+ 40 epochs (shipped recipe)	3	56.8$\pm$1.1	469	61.4%	77.1%
(non-compliant) 3-model ens.	1	52.0	—	—	—
(non-compliant) 6-model ens.	1	49.7	—	—	—

Table 1: Median/mean in km. Shipped row is mean $\pm$ sample s.d. of three seeds (55.60 / 57.12 / 57.71); **best observed 55.60**, but the submitted model uses the config’s default seed, which was not among the three — no seed was chosen on results. Geo-cell predecessor’s mean and rate columns were not retained. Ensembles are recorded to show the gain was real and rejected: the rules require one ≤ 5 M model.

Which half did the work? The -31 km row changes two things at once, so we separate them on fold 0: baseline (no BYOL, no country-aware, ep 6) 76.1 km; country-aware fine-tuning alone 70.0 km from baseline weights (ep 12) and 71.9 from a joint start (ep 24); adding the BYOL initialisation **53.6** (ep 22). Country-aware sampling buys ~ 6 km, the BYOL trunk ~ 16 km more — both contribute, BYOL more. Single fold, unequal epoch counts: indicative, not decisive. Extending 22 to 40 epochs is within the ± 1.1 km seed spread, so we do not claim it as an improvement.

5 Where the error comes from

Splitting each query into *located* (≤ 50 km), *mis-ranked* (a ≤ 50 km bank image was in the top-200 but not returned first) and *never retrieved* separates the two failure modes.

	located	mis-rank	not retr.	bank < 1 km
Pooled	49.1%	30.3%	20.7%	30.1%
Iceland	82.9%	15.2%	1.9%	57.6%
France	25.2%	32.7%	42.1%	14.5%
Germany	14.2%	40.6%	45.2%	9.1%

Table 2: One seed (the 55.60 km run), so pooled “located” is 49.1% rather than the 3-run mean 48.8% of Table 1. Last column: share of queries with *any* bank photo within 1 km. Full 12-country version in the repository README.

Germany (505 km median) and France (570 km) dominate what remains. They are 17% of the evaluation set; excluding them the pooled median falls from 55.6 to **31.7** km, and merely bringing them to Poland’s level would give 41.8 km. Ranking is the larger loss pooled,

GeoMatch: Single-Model Visual Retrieval Architecture

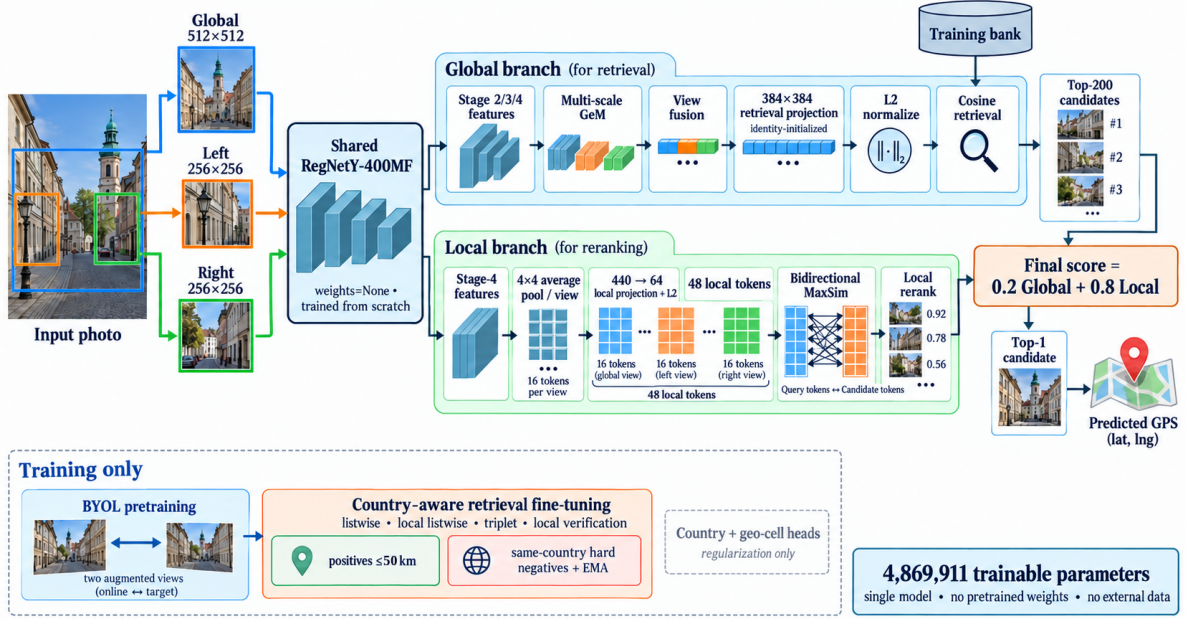


Figure 1: One from-scratch RegNetY-400MF trunk, shared across a global view and two crops, feeds a global-descriptor retrieval branch and a local-token late-interaction reranker. Classification heads regularise training and are unused at inference. 4,869,911 parameters.

but for these two nearly half of queries retrieve nothing nearby at all — both stages fail, so a better reranker could not have closed it.

Bank density reframes what retrieval is doing. Ranking the twelve countries by how often a bank photo lies within 1 km reproduces the performance ranking almost exactly (Spearman -0.89 against median error): Iceland 57.6%, Germany 9.1%. This does *not* explain the failure — the nearest German bank photo is still a median 8.8 km away, against a 505 km German median, so the encoder is lost, not starved. What it does say is that where the model looks strong it is often matching near-duplicates of the same place, and the countries where that crutch is absent are exactly the ones where the descriptor has to generalise and does not.

6 What did not work

Decode-time repairs on frozen descriptors (5-fold evidence). A learned candidate reranker was tried four times (linear, residual, pairwise, and one gated on a held-out inner fold) and degraded the crossfit every time; the inner-fold gate passed only because the probe fold was held out from the reranker but not from the *encoder*. Kernel-density consensus decoding gave 72 km against 57.7. Gating the shortlist on predicted country cost 4 km pooled (country accuracy is 68%, and a wrong gate is unrecoverable). Decoding from the auxiliary geo-cell head instead of retrieval gave DE 490 / FR 523 km against that encoder’s retrieval 534 / 551: two decoders reading one descriptor fail equally, which locates the problem upstream of the decoder — though it does not establish what the descriptor is missing.

Encoder variants (fold 0 only — weaker evidence). Adding a random-resized-crop to the global view was worse at every matched epoch (63.7 vs 58.1 km at epoch 32), consistent enough to call a backfire; it discards the

horizon and scene layout. Raising the local grid from 4×4 to 8×8 was *not* worse (58.3 vs 60.7 km at epoch 40) — we kept 4×4 as the simpler option, not because the variant failed; extending it to 48 epochs drifted to 61.8 km. Interpret both cautiously: the shipped recipe’s own fold-0 median varies by 5.3 km across epochs 20–40, so single-fold gaps of that size are not decisive.

7 Limitations

Germany and France are the dominant failure and we did not solve them. The evidence locates the problem in the representation rather than the decoder, but we never ran the experiment that would say *why* — e.g. probing whether the descriptor carries any within-country position signal. Also: the failed-variant evidence is single-fold and confounded by epoch count, and the figures are rebuilt from cached out-of-fold predictions that are not in the repository, so they cannot be regenerated from a clean clone.

8 Submission

The recipe retrained once on all 11,758 images with no held-out fold, from an existing BYOL trunk, using the config’s default seed: 4,869,911 parameters, **67 min of fine-tuning** on one RTX 4000 Ada (excluding BYOL pretraining, ≈ 2 h 40 min, reused rather than re-run), plus ≈ 5 min to write `predictions.csv`.

- [1] Radosavovic et al. Designing Network Design Spaces. CVPR 2020.
- [2] Radenović, Tolias, Chum. Fine-tuning CNN Image Retrieval with No Human Annotation. TPAMI 2019.
- [3] Khattab, Zaharia. ColBERT: Late Interaction over BERT. SIGIR 2020.
- [4] Grill et al. Bootstrap Your Own Latent. NeurIPS 2020.